Deterministic Connectome-to-Artix-7 Compilation: A Normative Bit-Exact Contract, Canonical Lowering, and Audited Pre-Board Closure

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Abstract

We present a contract-first connectome-to-hardware artifact package for deterministic compilation from a vendored *C. elegans* structural connectome to synthesizable hardware targeting AMD Artix-7 FPGAs. The central claim is intentionally narrow and reviewer-facing: this release closes the pre-board evidence chain, not board measurement. The package combines (i) a normative fixed-point tick semantics (`nema.tick.v0.1`) defined as a discrete-time leaky integrator with tanh-LUT activation, current-based chemical synapses, symmetric ohmic gap junctions, optional `delayTicks` via ring-buffer snapshot reads, and fixed-point RNE+SAT; (ii) a deterministic lowering path from a minimal declarative description to canonical IR, HLS-oriented C++, and reference RTL; (iii) end-to-end bit-exact verification via per-tick SHA-256 digest traces computed only from serialized `V_raw` (signed int16 little-endian, node-index order) across the golden software model, vendor HLS `csim/cosim`, and released implementation artifacts; and (iv) an audited Artix-7 pre-board evaluation on part `xc7a200tsbg484-1` including post-route QoR and SAIF-guided power estimation. For the canonical structural benchmark B3, the release closes HLS digest parity, post-route implementation, and post-route power estimation while explicitly labeling all power and energy numbers as `ESTIMATED_PRE_BOARD_ONLY`. The resulting artifact is suitable for arXiv release and external review because every quantitative claim is backed by generated CSV/JSON tables, report files, and hashes.

1 Introduction

The *C. elegans* hermaphrodite connectome remains one of the few biologically grounded network datasets that is both historically influential and openly reusable. The classical Varshney dataset reports a near-complete neuronal wiring diagram and distinguishes chemical synapses from gap junctions [1, 2]. In parallel, FPGA-oriented spiking frameworks and neuromorphic intermediate representations have matured substantially, spanning interoperable IRs, simulator stacks, and open accelerator generation flows [3, 4, 5, 6, 7, 8]. What remains uncommon is a reviewer-proof contract layer: a deterministic and auditable path from a biological graph to hardware-facing artifacts with explicit fixed-point semantics, end-to-end digest equivalence, reproducibility manifests, and a clear separation between estimated pre-board quantities and future on-board measurements.

This paper is the integrated version of the project narrative. Earlier revisions already established the semantic and reproducibility logic, but the last short release note compressed that logic into a three-page status wrapper. Here we restore the full research arc and update it with the now-closed

pre-board Artix-7 evidence chain. The result is not a claim of finished board deployment. It is a claim that the pre-board methodology is complete, traceable, and reproducible.

Contributions.

1. A normative fixed-point and tick-level semantics (`nema.tick.v0.1`) defined as a discrete-time leaky integrator with tanh-LUT activation, current-based chemical synapses, symmetric ohmic gap junctions, delayed snapshot reads via `delayTicks`, Euler update, exact RNE+SAT behavior, and bit-exactness defined from `V_raw`-only digest traces.

2. A deterministic lowering contract from declarative connectome-plus-sidecar input to canonical IR, HLS-oriented C++, and reference RTL, with stable serialization and reproducible hashes.

3. A canonical executable benchmark derived from the vendored Varshney/OpenWorm raw spreadsheet, plus strict identity tracking from raw dataset through sidecar, manifest, IR, HLS, and implementation evidence.

4. End-to-end bit-exact verification across the golden software model, vendor HLS `csim`/`cosim`, and post-route implementation artifacts for the required B1 and B3 benches.

5. An audited pre-board Artix-7 closure on `xc7a200tsbg484-1`: post-route QoR, SAIF-guided power, and derived throughput/energy metrics, all explicitly labeled as pre-board estimates rather than on-board measurements.

2 Related work and research gap

The artifact sits at the intersection of neuromorphic IRs, FPGA spiking accelerators, and connectome-driven neural simulation. NIR explicitly frames interoperable neuromorphic computation as a shared representation problem [3]. Lava offers an open software stack for mapping neuro-inspired applications to neuromorphic hardware [4]. FINN and hls4ml are influential open workflows for FPGA compilation, but they focus on quantized inference and ML deployment rather than biological connectome contracts [5, 6]. Spiker+ and NeuroCoreX are closer in spirit because they target spiking neural networks on FPGA and are open source, yet their emphasis is accelerator generation and flexible SNN experimentation, not deterministic connectome lowering with digest-visible semantic obligations [7, 8]. On the biological side, BAAIWorm and NeuroSimWorm show the continued vitality of whole-organism or multisensory *C. elegans* modeling, but they do not solve the same compilation problem [9, 10].

The gap addressed here is therefore specific: open-source availability *plus* deterministic lowering *plus* normative semantics *plus* bit-exact traceability *plus* synthesizable Artix-7 evidence. We also take seriously the observation from recent formal-HLS work that transformations above RTL can be subtle enough to justify machine-checkable reasoning and translation validation [11]. That motivates the contract-first stance adopted throughout this paper.

3 Normative semantics

The binding semantics are defined in the repository normative specification and summarized here.

3.1 Numeric model

All dynamic values use signed two's-complement fixed point with width W and F fractional bits. Source-side rationals and decimals are lowered by round-to-nearest, ties-to-even (RNE), followed by saturation. Addition and subtraction use an unbounded intermediate domain and saturate on commit. Multiplication computes the full product, rescales by 2^F using RNE, and saturates. These rules are normative rather than descriptive: if an implementation deviates here, it is not conformant even if it remains numerically close.

3.2 Architectural state and tick rule

In `nema.tick.v0.1`, the persistent per-node state is voltage V in signed Q8.8 (int16 raw). Per tick, the simulator derives activation A from a tanh LUT (Q8.8) and uses temporary current accumulators I_{chem} and I_{gap} in signed Q12.8 (20-bit raw domain). Delay handling uses `delayTicks` by reading prior voltage/activation snapshots from ring buffers; there are no digest-visible delay-bucket structures. Each tick follows a strict snapshot rule: (1) snapshot all voltages; (2) compute $A_i = \text{tanhLUT}(V_i^{snapshot})$; (3) accumulate chemical current over directed edges with $msg = \text{quantize}_{Q12.8}(coeff \cdot A_{pre})$ and saturating adds; (4) accumulate gap current once per undirected pair (a, b) with $msg = \text{quantize}_{Q12.8}(g \cdot (V_b - V_a))$ and saturating $\pm msg$ updates; (5) compute Euler update from snapshot currents, $V_{next} = \text{saturate}_{Q8.8}(V^{snapshot} + \text{quantize}_{Q8.8}((dt/\tau_m) \cdot (I_{chem} + I_{gap})))$.

3.3 Bit-exactness

End-to-end equality is defined operationally rather than approximately. At every observed tick, only V raw values are serialized in node-index order as signed int16 little-endian and hashed with SHA-256. Two implementations are bit-exact on a benchmark iff every observed tick yields the same digest.

4 Deterministic compilation pipeline

The pipeline is intentionally narrow:

connectome + sidecar → canonical IR → golden CPU model → HLS-oriented C++ → reference RTL.

Determinism is enforced by canonical ordering of neurons and edges, stable JSON serialization, versioned manifests, and explicit dataset provenance fields. No pass is allowed to use hidden randomization. The executable benchmark released here is derived from a vendored raw spreadsheet (`NeuronConnectFormatted.xlsx`) together with a deterministic importer and sidecar. The canonical B3 manifest in the final release resolves to `B3_varshney_exec_expanded_gap_300_5824`: 300 nodes, 3638 directed chemical edges, and 1093 undirected gap pairs (2186 directed gap entries in the raw directed representation), for 5824 total directed edges after deterministic filtering.

Two details matter for reproducibility. First, the compiler records dataset and sidecar digests in manifests and IR. Second, the benchmark identity itself is treated as part of the conformance contract; an alias to a different graph is not sufficient for gate closure. This is exactly the issue that kept earlier HLS closure attempts open until the canonical B3 identity was materialized and propagated through the flow.

bench	key	ticks	gold	csim	cosim	g=csim	g=cosim	csim=cosim	runner	csim	cosim
b1_small	b1_small	20	20	20	20	Y	Y	Y	0	PASS	PASS
b3_canonical	b3_varshney_exec_expanded_gap_300_5824	20	20	20	20	Y	Y	Y	0	PASS	PASS

Table 1: Strict HLS digest parity for the required benches. B3 denotes the canonical structural benchmark `B3_varshney_exec_expanded_gap_300_5824`.

Benchmark	ImplOK	LUT	FF	BRAM	DSP	WNS(ns)	TNS(ns)	Period(ns)	Fmax(MHz)	Part
b1_small	yes	474	323	0	0	-1.743000	-92.501000	5	148.301943	xc7a200tsbg484-1
b3_varshney_exec_expanded_gap_300_5824	yes	2734	2862	7	2	0.356000	0	5	215.331611	xc7a200tsbg484-1

Table 2: Post-route QoR on Artix-7 part `xc7a200tsbg484-1`.

5 Reference microarchitecture and hardware flow

The intended hardware organization is a deterministic CSR-based, time-multiplexed engine. Chemical edges are stored in CSR by post-neuron, gap junctions are stored as canonical endpoint-pair tables, and delayed reads are supported via BRAM-friendly ring buffers for prior voltage/activation snapshots. The hardware schedule mirrors the normative tick rule exactly: snapshot voltages, LUT activations, chemical accumulation, gap-pair accumulation, then Euler voltage update.

The representative FPGA target is the cost-optimized Artix-7 family [15]. In Vivado, the concrete part used in the final release is `xc7a200tsbg484-1`. Pre-board closure follows the vendor-supported flow: HLS `csim`/`cosim` for behavioral conformance [12], post-route implementation for timing and resource evidence, and SAIF-guided power using xsim plus `read_saif`/`report_power` on the implemented design [13, 14].

6 Verification and audited closure

The verification stack is layered deliberately.

6.1 Golden model and digest traces

The software golden model is the normative reference. Every benchmark emits per-tick digest traces, and those traces are the comparison contract used throughout the pipeline.

6.2 Vendor HLS parity

The final release closes HLS parity for both required benches. Table 1 shows that the golden model, vendor `csim`, and vendor `cosim` agree at every observed tick for B1 and canonical B3. This is the evidence that closes G1b.

6.3 Implementation evidence

Table 2 summarizes post-route QoR on `xc7a200tsbg484-1`. Both B1 and B3 have real post-route evidence, which closes G1c. B1 does not meet a 200 MHz target; its post-route WNS is -1.743 ns, corresponding to an estimated 148.30 MHz. B3, by contrast, closes timing at 200 MHz with positive slack and an estimated 215.33 MHz fmax headroom. The important point is not that every benchmark meets the same clock target, but that the Artix-7 implementation evidence now exists and is traceably bound to the correct part.

Benchmark	Ticks	SAIF Tot(W)	SAIF Dyn(W)	SAIF Stat(W)	Match	read_saif	power
b1_small	50	0.139000	0.008000	0.131000	1059/1329	PASS	PASS
b3_varshney_exec_expanded_gap_300_5824	10	0.163000	0.031000	0.132000	4147/8683	PASS	PASS

Table 3: SAIF-guided post-route power estimates from functional xsim simulation.

Bench	LUT	FF	BRAM	DSP	WNS(ns)	f_run(MHz)	cycles/tick	ticks/s	P(W)	E/tick(mJ)
b1_small	474	323	0	0	-1.743000	148.301943	-	-	0.139000	-
b3_varshney_exec_expanded_gap_300_5824	2734	2862	7	2	0.356000	200	15008	13326.226013	0.163000	0.012232

Table 4: Derived metrics for canonical B3 using the final SAIF-guided power selection.

6.4 SAIF-guided post-route power

The final release replaces the earlier synthetic clock-only activity with functional post-route xsim SAIF from tick-driven benchmark testbenches. Table 3 reports the SAIF-guided power values used for the final release. This closes G1d. The values remain `ESTIMATED_PRE_BOARD_ONLY`: they are vector-based post-route estimates, not silicon measurements.

6.5 Derived throughput and energy

For B3, the release also derives cycles per tick from the HLS latency report and combines that with the implementation frequency and SAIF-guided total power to estimate throughput and energy per tick. The resulting value is 0.012232 mJ/tick (about 12.232 µJ/tick) at 200 MHz. Table 4 reports the release-selected derived metrics.

7 Evaluation summary

Three points matter most for interpretation. First, the benchmark used to close the structural gate is the canonical 300-neuron, 5824-edge B3 instance generated by the deterministic importer, not the earlier provisional aliases used during debugging. Second, timing closure is benchmark dependent: B3 meets the 200 MHz target on the selected Artix-7 part, whereas B1 does not. Third, power results are now based on functional SAIF rather than synthetic activity, which materially improves representativity, but they remain estimates until board telemetry exists.

8 Threats to validity

The largest remaining limitation is unchanged: there is still no board measurement in this release. All power and energy values are pre-board estimates. SAIF-guided post-route power is stronger than vectorless power and stronger than a clock-only harness, but it still depends on the realism of the chosen testbench activity window [14, 13]. A second limitation is toolchain dependence: source-level determinism does not imply that QoR is invariant across vendor versions or host environments. A third limitation is model-instantiation scope. The release closes a deterministic executable benchmark instance derived from a public structural dataset and explicit sidecar rules; it does not claim that all biological parameters are uniquely identifiable from structure alone.

9 Reproducibility and release structure

The release is intended to be reviewer-checkable. The final bundle contains the paper source, the final PDF, selected evidence tables, the artifact bundle, and SHA-256 hashes. The machine-readable

release status lives in `FINAL_STATUS.json`, while gate closure rationale lives in `GATE_STATUS.md`. The selected tables for HLS parity, QoR, power, and derived metrics are all generated artifacts rather than manually edited tables. Code and artifacts are publicly available at https://github.com/tangodelta217/nema-connectome-artix7.

For arXiv, the correct upload unit is the LaTeX source package rather than a stand-alone PDF. The release already contains `paper.tex`, `refs.bib`, and the generated tables needed to rebuild the PDF.

10 Conclusion

The project now reaches a clean pre-board stopping point. The semantic contract, deterministic lowering path, canonical benchmark identity, HLS digest parity, Artix-7 post-route implementation, and SAIF-guided post-route power are all closed with auditable evidence. What remains open is exactly what should remain open without hardware: on-board telemetry and physical-board validation. That division is a strength rather than a weakness, because it lets reviewers see which claims are complete today and which claims are explicitly future work.

Future work. The next milestone is straightforward: execute the predefined board-measurement protocol on a physical Artix-7 board, correlate measured rail or wall power with the present SAIF-guided estimates, and decide whether additional benchmark-side activity modeling is needed to tighten the gap between pre-board and in-board evidence.

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